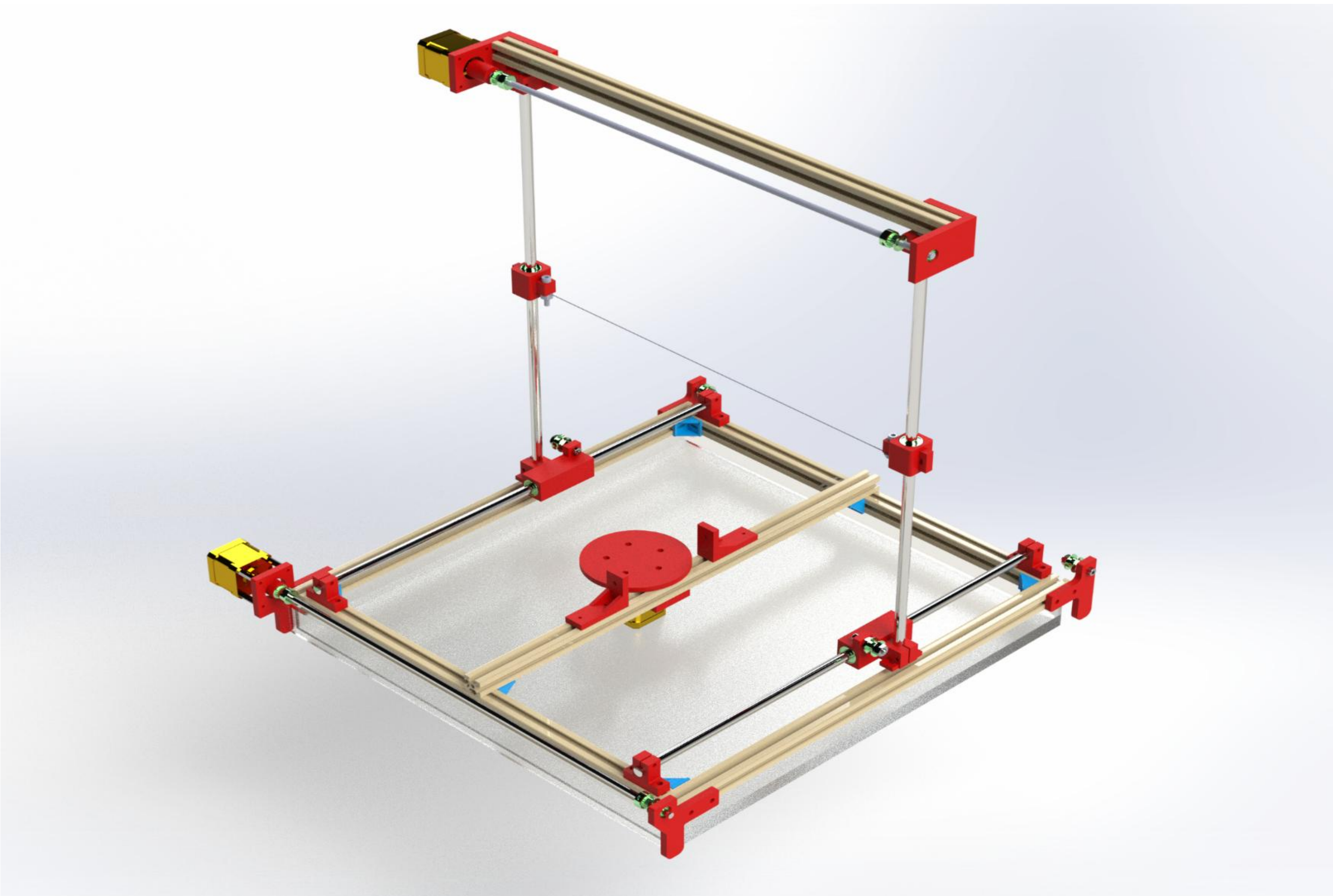
ULTRA LOW-COST AUTOMATED EPS CUTTER



Submission Date: 21th July, 2022

Department of Industrial and Production Engineering (IPE),

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Course Title: Product Design I (Sessional)
Course Code: IPE 304
Product Name: Ultra Low-Cost Automated EPS Cutter

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Forwarding Letter

July 21, 2022

Md Muzahid Khan, Lecturer

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Department of Industrial and Production Engineering

Military Institute of Science and Technology

Subject: A report on "Ultra Low-Cost Automated EPS Cutter".

Dear Teachers,

With due respect, we would like to submit you a report on " Ultra Low-Cost Automated EPS Cutter ".

Day by day we are depending on machines to get our chores done. Machines can do work with high accuracy and efficiently. But we still use manual foam cutting which does not provide us with accurate shape. The process of manufacturing the plastic foam is the casting and extruding, so the perfect way to cut these types of foam is hotwire cutting. That’s why we are going to build “Ultra Low-Cost Automated EPS Cutter”, a CNC machine which will cut foam using hot wire and can cut intricate shapes. The design methodology of the 'Eco-friendly Waste Collector' is presented in this report.

We would like to thank you for helping us with suggestions and advice for completing this report. There might have been some errors and lacking in our report due to a shortage of time. We earnestly apologize for such mistakes on our part. Therefore, we request you to kindly consider such cases and oblige thereby.

Sincerely Yours,

Md. Soad Solaiman

Level-3, Term-1

Department of Industrial and Production Engineering, MIST

Preface

The word 'Design' means not only to draw but to make a plan, a purposeful outline of anything usable. According to Steve Jobs, "Design is not just what it looks like and feels like. Design is how it works." So, in engineering, the design combines the overall process of planning with the application.

In our Product Design I sessional, we have learned how to design a new product or to improve the design of an existing product. In the Product Design-II sessional course (which will be covered in the next term), we will manufacture it. So, every step of a product development process will be performed.

After consulting with our course teachers, 'Ultra Low-Cost Automated EPS Cutter' was selected as our sessional course project. ' Ultra Low-Cost Automated EPS Cutter ' is a CNC machine which will cut foam using hot wire and can cut intricate shapes.

The main aim of this project was to learn various theoretical Product Design Course concepts practically by working on this design development process. This will also help us to design and improve more complex products in the future.

Engineering is not only about theoretical knowledge but also about applying them in real-life situations. This project is extremely important as it is the practical result of our theoretical concepts. This is the overall outcome of what we have learned in Product Design-I theory and sessional courses.

Summary

The development of industry is growing continuously. As an example, making decorations from Styrofoam Decoration can be used in various things, placards, art work, making safety goods on shipping, and others. The use of CNC Wire Cutter in making decoration will shorten the time needed. Our target is to manufacture a product which will effectively increase work efficiency. To make the foam cutting system more acceptable, we try to identify customer's needs. Analyzing them, we have changed some features which was very helpful. We have made a prototype of the machine, have selected materials, machining processes for giving a shape to that prototype and analysis the cost to evaluate the cost and benefit. We did all these things by following certain steps so that every step done perfectly and our product can give a strong performance in the competitive world.

Acknowledgement

The main objective of the Product Design-I sessional is to apply our concepts & knowledge into practical life. This was a task that was quite difficult. Still, it'd had been more difficult if we had not got some continuous help from some generous people. Their continuous help made our journey easy in this product design process.

Firstly, we would like to impart our earnest gratitude to our respected course teachers. They have helped us tremendously in developing our concepts & completing our project in many difficult situations which would be quite tough for us to handle.

We also want to convey our holy gratitude to all the people who have given us all necessary information about the development process.

Finally, we would like to thank our group members, classmates & our beloved seniors for their continuous information & help.

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Chapter 1
Introduction

1.1 Product Design

Product design is a set of strategic & tactical activities beginning from the idea generation up to the commercialization of the product. It is a detailed process of imagining, creating, and iterating products that either solves the problems of the customers or address the specific needs in the market. Product designers work to optimize the user experience in the solutions and help their brands by making products feasible for longer-term business.

Actually, the design process includes various forms of sketching and prototyping. However, it is about establishing a bridge between the user and the environment.

1.2 Our Product

Day by day we are depending on machines to get our chores done. Machines can do work with high accuracy and efficiently. But we still use manual foam cutting which does not provide us with accurate shape. The process of manufacturing the plastic foam is the casting and extruding, so the perfect way to cut these types of foam is hotwire cutting. That's why we are going to build "**Ultra Low-Cost Automated EPS Cutter**", a CNC machine which will cut foam using hot wire and can cut intricate shapes.

1.3 Planned Features

The main objective is to make a low-cost CNC machine which can machine foam. The features we want to include in our product are as follows:

a. Fully software based automatic cutting.
b. Dust free cutting.
c. No noise pollution.
d. Simple operation and maintenance.
e. Can operate complex design operations.
f. Solar panel is also used as a renewable power source.

1.4 What is the purpose of making this product?

In the manual foam cutting process, there may occur many errors. The drawbacks of manual cutting process are:

1. To cut foam in particular shape there is need of precise templates from a hard material (like ply, acrylic sheet), they should be firmly attached to the flat surface of the foam.
2. In aircraft design the weight is an important aspect, but making a hollow cross section is a bit hard.
3. Manual flow of hot wire over foam is not smooth, which leads to uneven surface finish.

To get rid of these drawbacks and to get good quality cuts we use **Ultra Low-Cost Automated EPS Cutter**.

Chapter 2
Literature Review

The development of industry is growing continuously. As an example, making decorations from Styrofoam Decoration can be used in various things, placards, art work, making safety goods on shipping, and others. The use of CNC Wire Cutter in making decoration will shorten the time needed [1].

The **CNC machine** comprises the mini computer or the microcomputer that acts as the controller unit of the machine. CNC Machining is a process used in the manufacturing sector that involves the use of computers to control machine tools automatically. A computer program is customized for an object and the machines are programmed with CNC machining language (called G-code) that essentially controls all features like feed rate, coordination, location and speeds. [2]

G-Code is one of the computer code languages that are used to instruct CNC machining devices what motions they need to perform such as work coordinates, canned cycles, and multiple repetitive cycles.[3]

Expanded polystyrene (EPS) is a rigid, closed cell, thermoplastic foam that comprises about 2% polystyrene and 98% air. This foam is light, usually 1530 kg m3, and inexpensive. Extruded polystyrene (XPS) foam is chemically the same as Expanded foam (EPS), but it is manufactured using a different process.[4]

We will use **CAD** software to design our product. Computer-aided design (CAD) software allows designers and manufacturers to produce a model with technical specifications, such as dimensions and geometries, for producing the part or product.[5]

There are several mechanisms for the cutting of plastic foam; they can be divided into three-basic models or the mixing among them:

- **Thermal cutting:** In this mode, the plastic foam is vaporized or just melting in front of the wire cutter but without touching between them. See Figure a.
- **Thermo-mechanical cutting:** this mode is a mixing between the shearing force and the melting of material to perfume the cutting and the hot-tool in contact with the material. See Figure b.
- **Mechanical cutting:** this mode depends on the shearing because it happens when the temperature of the tool is below the melting point of the foam [6] as shown in Figure c.

From this cutting process we will use **Thermomechanical** cutting process in our product.[6]

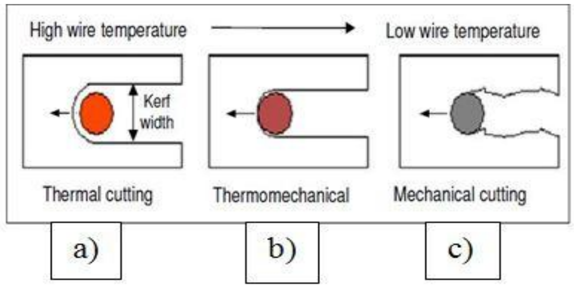


Fig. 1. Cutting mechanisms

In order to cut foam more accurately, it requires a lot of skills and manpower to achieve the desired results. This research is an effort to build a system that serves the needs we want which is to be faster and more accurate.

Conclusion

The Automated EPS Cutting Machine is an effective method to prepare foam molds of multifarious shapes. Here, the hot wire has High Durability with a thin shape. Besides, as the CNC method is used here, the speed of the machine can be adjusted with various feed rates as per the industry. This hardware setup with a combination of G-code will give better accuracy and reduce the workload. Some of the factors that can also be considered during the design of the machine are safety, power requirement, compactness, efficient cost production, material selection and ease of fabrication.

Chapter 3
Understanding Customer Needs, Gathering and Prioritizing Data

3.1 Introduction

The main key to run a successful business is to meet customer needs. Business products which understand the customer needs are likely to outperform their competitors and also it's the easiest way to position your product smartly in the market. To identify these needs and expectations, we may both ask and listen to the right questions and present the right product, services and solutions to meet customer needs which will help to grow the business. This survey mainly focuses on satisfying the customers' need for producing a product of superior value as well as for determining the longevity of our business.

There are many companies, event managements, different types of clubs such as Arts and crafts club, Science club, Cultural club, Robotics Club, etc and various organizations use foam when any kind of occasion occurs. To make the cutting process easier and reduce time, cost and cut the foam more effectively with high accuracy, we are going to build an "**Ultra Low-Cost Automated EPS Cutter**". For our product, we have identified our targeted customers:

- Event management companies.
- NGO's.
- Social activities club.
- Educational organizations.
- Architectural firm.
- Aerospace industries.

3.2 Areas and Locations of Survey

- Mirpur, Dhaka
- Nilkhet ,Dhaka
- Uttara Area, Dhaka
- Shahbagh, Dhaka
- Dhaka Cantonment

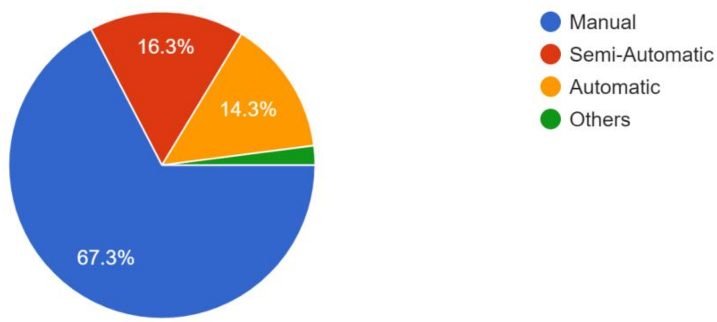
3.3 Medium of Survey

- Online survey through Google forms
- Face to face interview
- Interview over phone

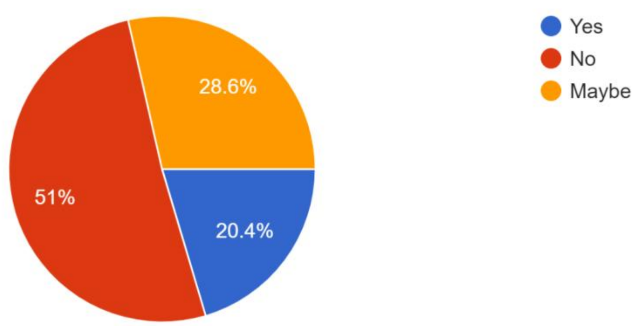
We did our survey of 49 people. Based on their variable answers, we made our survey results with the following pie charts. For calculating percentage, fractions were rounded up to the nearest integer number.

3.4 Survey Result

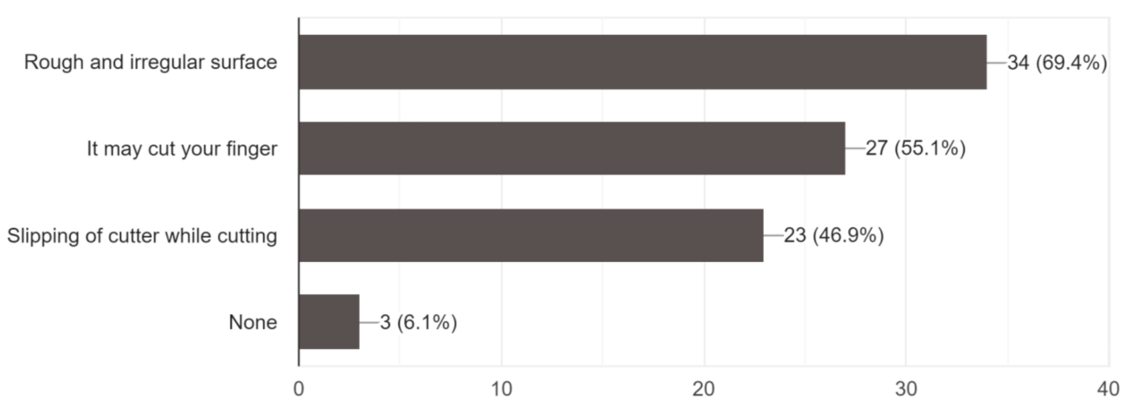
1. What type of cutter do you use while cutting foam?
49 responses



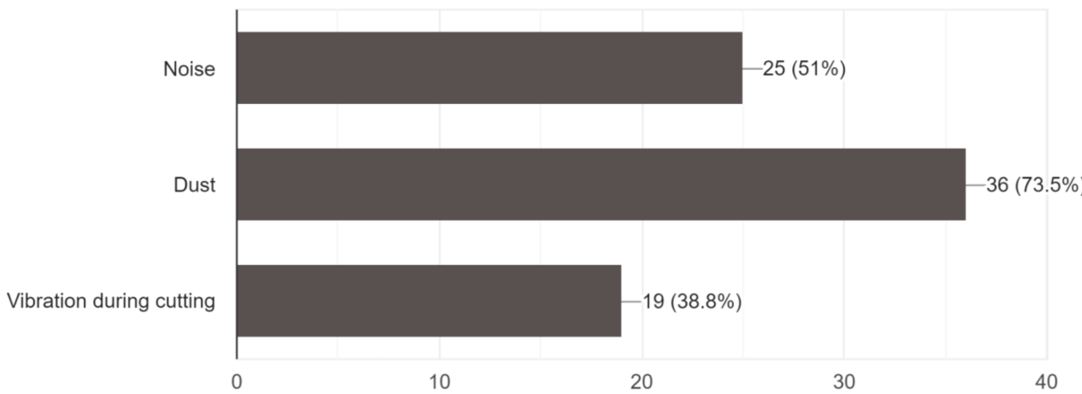
2. Could you cut your desired shape by using a knife?
49 responses



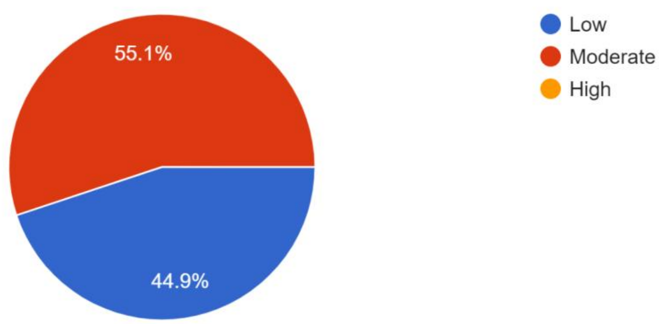
3. What makes you uncomfortable during cutting foam manually? (You can choose multiple option)
49 responses



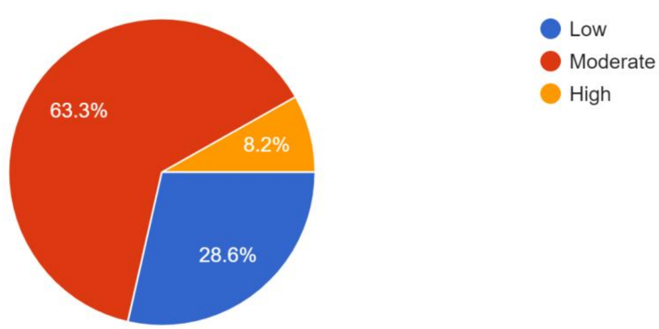
4. What issue makes you irritated when you try to cut a foam?(You can choose multiple option)
49 responses



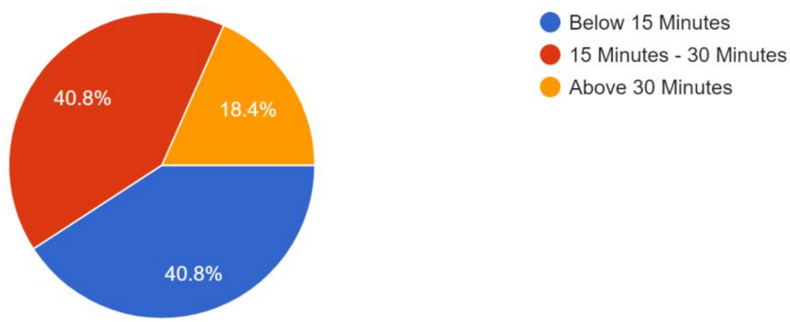
5. What do you think about the accuracy of manual foam cutting?
49 responses



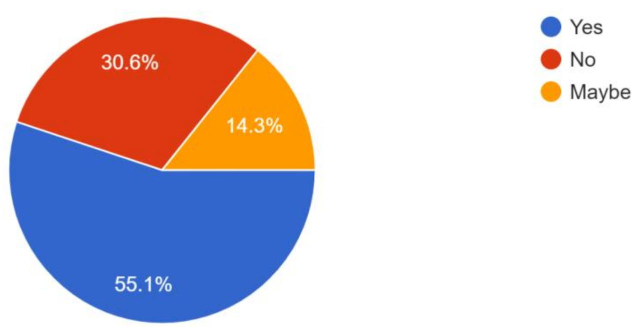
6. What is the level of foam cutting skill of you or your organization?
49 responses



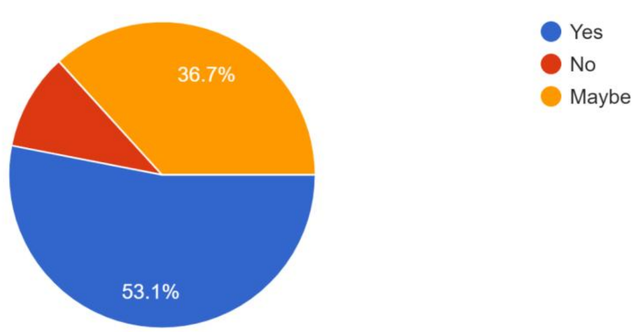
7. How much average time is taken by manual foam cutter, If you cut 'A' letter in regular shape?
49 responses



8. Are you or your organization familiar with designing software?
49 responses

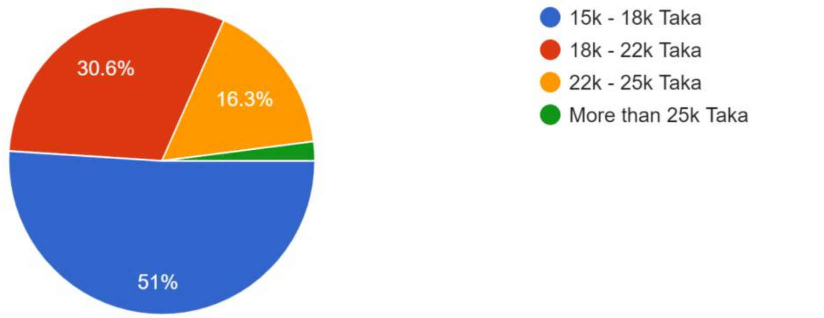


9. If an automated foam cutter is made, would you prefer to buy it?
49 responses



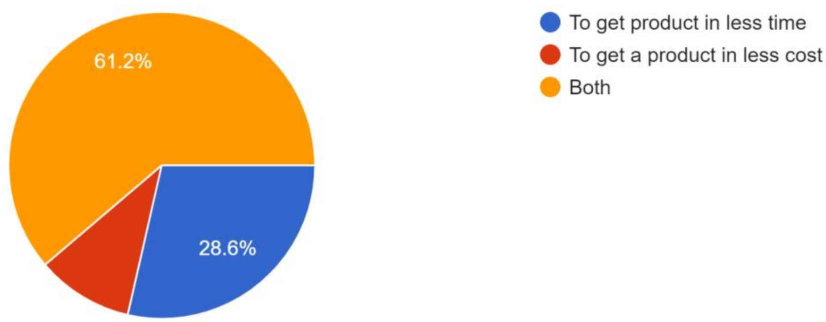
10. What should be your expected price?(You can choose multiple option)

49 responses



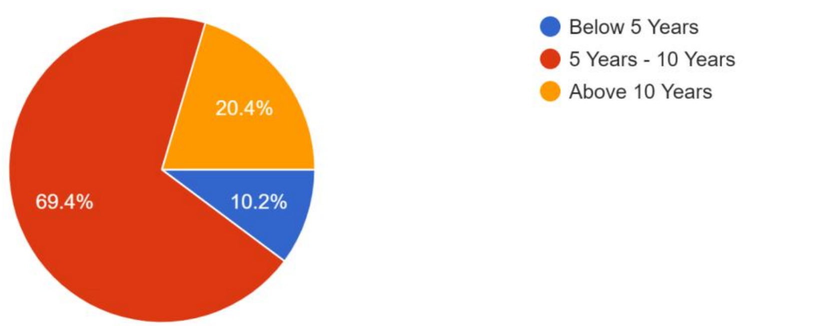
11. What do you want from this machine?

49 responses

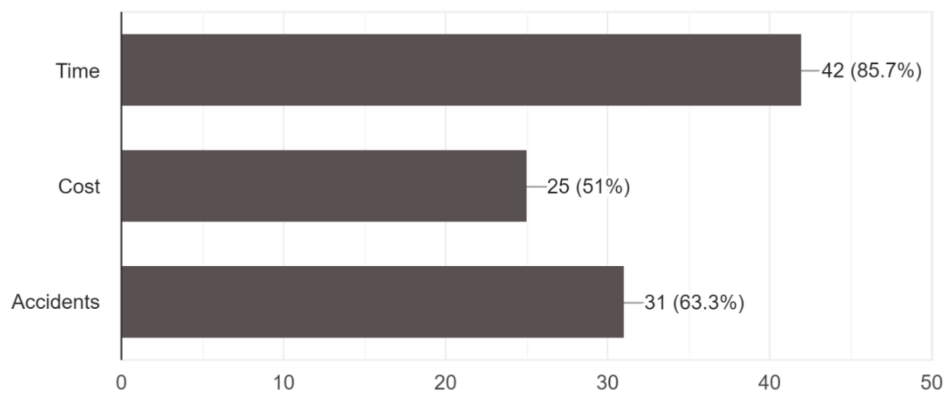


12. What should be the lifetime of this product in your opinion?

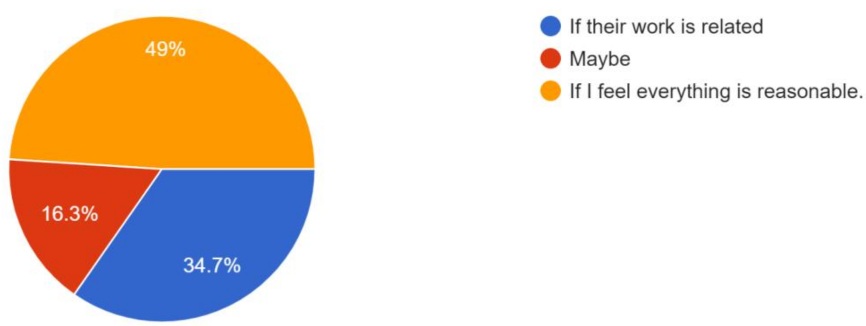
49 responses



13. What should be most likely to be reduced while using this product in your opinion?(You can choose multiple option)
49 responses



14. Would you like to recommend others to buy this product?
49 responses



15. What do you suggest to modify this product?
  - Don't limit it to a cork sheet.
  - Make this machine able to shape any material. As a result, casting, welding and many complicated processes may be avoided.
  - I think if the product can work precisely and has a timer which will provide me the time needed for finishing my design, it will be more reliable for me.
  - Not only focusing on foam you should consider cutting wood and plastic.
  - It will help to inspect how the product will be shaped before mass production with effectively and accurately in less time.
  - Add safety features if possible.
  - Try to reduce the operation time as much as possible. It would be helpful if the required maintenance for this machine is very less.
  - Try to make the product cost efficient and energy efficient including the safety measures.
  - The idea is itself perfection. So my suggestion will be the proper implementation of the current idea.
  - It should be effective and free from physical harm to humans.
  - By ensuring a safety system you can modify your product.
  - Make it light weight.

3.5 Customers' Requirements

- Eco-friendly
- Time-saving
- Safe machining process
- Reasonable price
- User friendly
- Durability
- High accuracy
- Light weight
- Power Saving

3.6 Conclusion

In this survey, our major purpose was to find what people really think about our **Ultra Low-Cost Automated EPS Cutter**. We prepared questionnaires for our targeted customers to collect information about what they actually want. After the survey, we are able to determine the requirements of the customers. Thus, how we know which aspect is the most prioritized. Therefore, we are going to design our product keeping these demands in mind very carefully and optimize it in all possible ways.

Chapter 4
Incorporating Voice of Customers in Product Design with QFD

4.1 Introduction

To run a successful business, we need to produce our product according to our targeted customer needs. Customer needs should be expressed in the language of the customer in order to provide specific guidance about how to design and manufacture a product. Understanding the design problem is a must when we try to develop a quality product. A development team establishes a set of specifications, which spell out in precise, measurable detail what the product has to do to be commercially successful.

Quality Function Deployment (QFD) is a process and set of tools used to effectively define customer requirements and convert them into detailed engineering specifications and plans to produce the products that fulfill those requirements. User experiences confirm that QFD can facilitate the following:

- Reduced product development cycle time.
- To transform qualitative user demands into quantitative parameters.
- Developing the specifications.
- Determining targets to work toward.
- Improve customer satisfaction.
- Increased competitiveness.

Voice of the Customer (VoC) is a term that describes customer's feedback about their experiences with and expectations for your products or services. It focuses on customer needs, expectations, understandings, and product improvement. The voice of the customer is captured in a variety of ways:

- Direct discussion or interviews
- Surveys
- Customer provided specifications
- Direct observation
- Field reports.

4.2 House of Quality

House of Quality (HOQ) can be defined as the most convenient, easy, and simple tool used to convert the customer needs into technical descriptors for the firm. House of Quality is actually a matrix and is also termed as Quality Matrix. It is the structure in which information is organized, which is used by QFD. There are 2 best practices while thinking about House of Quality. Those are -

- Thinking quantitative, not just qualitative.
- Thinking outside the box, but being realistic.

Purpose of House of Quality -

- Understanding customer needs and desires.
- Translating Customer Desires into Goals & technicalities.
- Understanding the priorities of the Customer.
- Allocate resources.

4.3 Tools (HOQ) of Quality Function Deployment (QFD):

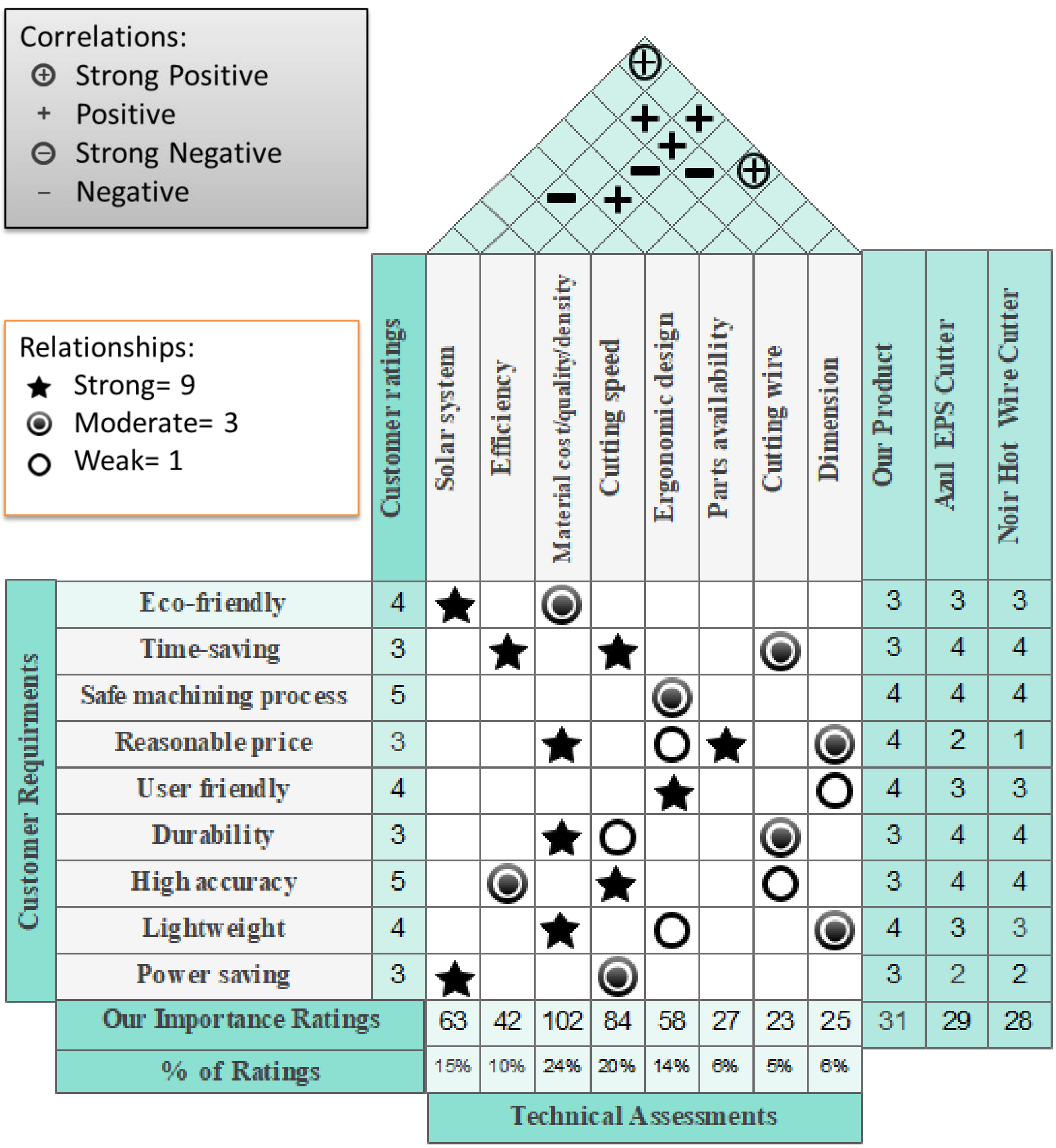


Fig: House of Quality for Ultra Low-cost Automated EPS Cutter

4.3.1 Relationship between Different Customer Needs and Technical Requirements

Customer Requirements	**Technical Requirements**	**Relationship**	**Explanation**
Eco-friendly	Solar system	Strong	Solar System is a renewable energy. It has no exaggerated output.
	Material type	Moderate	Reusable and biodegradable materials are eco-friendly.
Time-saving	Efficiency	Strong	Higher efficient machines can save a lot of time.
	Cutting speed	Strong	More the cutting speed, the time will save more.
	Cutting wire	Moderate	Higher quality cutting wire can save time during cutting.
Safe machining process	Ergonomic design	Moderate	Ergonomic design can make the machining process safe.
Reasonable price	Material cost	Weak	The higher the material cost, the more the price of the product.
	Parts availability	Strong	Parts availability makes the product price less.
	Dimension	Strong	To make the product larger will increase the cost.
	Ergonomic design	Moderate	To make the design

			ergonomic, the price will little increase.
User-friendly	Dimension	Weak	Suitable dimension makes the product user friendly.
	Ergonomic design	Strong	Ergonomic design makes the product user friendly.
Durability	Material quality	Strong	Higher quality material increases the product life-time.
	Cutting Speed	Weak	High cutting speed for a long time creates stress on the machine Thus reduces the product life.
	Cutting wire	Moderate	Higher quality Higher wire reduces erosion of wire.
High accuracy	Efficiency	Moderate	Higher efficient machines can provide high accuracy.
	Cutting Speed	Strong	Higher cutting speed provides lower accuracy and vice versa.
	Cutting wire	Weak	High quality cutting wire will provide high accuracy.
Light weight	Material density	Strong	Higher density material will increase product weight.
	Ergonomic design	Weak	To make the product ergonomic it’s weight is slightly compromised.

	Dimension	Moderate	Large product will increase product weight.
Power saving	Solar system	Strong	Using solar with battery will save the power more
	Cutting Speed	Weak	Lower cutting speed will consume less power.

4.3.2 Relationship Among Different Technical Requirements

Relationship Between	**Relation**	**Explanation**
Solar system and dimension	Strong positive	Attaching solar panel makes the product larger.
Efficiency and cutting speed	Negative	Higher cutting speed can't get time to melt the foam. Thus, friction happens more and efficiency decreases.
Efficiency and cutting wire	Positive	Higher quality cutting wire makes the cutting more efficient.
Material quality and ergonomic design	Positive	To make the product ergonomic, material quality has to increase.
Material cost and parts availability	Negative	Availability of parts makes the product cost least.
Material cost and cutting wire	Positive	Material cost increases to ensure higher quality cutting wire.
Material cost and dimension	Positive	Large Dimension of product increases material cost.
Cutting speed and cutting wire	Negative	Higher cutting speed erodes more cutting wire.
Ergonomic design and dimension	Strong positive	To make the product ergonomic, dimension has to be sacrificed.

4.4 Priority List of Technical Requirements

Order	Requirements	Weightage
1	Material cost/quality/density	24%
2	Cutting speed	20%
3	Solar system	15%
4	Ergonomic design	14%
5	Efficiency	10%
6	Dimension	6%
7	Parts availability	6%
8	Cutting wire	5%

4.5 Conclusion

Quality Function Deployment particularly the “House of quality” is an effective approach in ensuring customer expectations are used in driving the design process of a product. It is not used only for bringing the new product closer to the intended target, but also it reduces development cycle time and cost in the process. Using the QFD method we figured out the technical aspect of each customer’s requirement. It also enables us to design the product so that everything is in place to achieve the desired outcome.

Chapter 5
Functional Decomposition

5.1 Introduction

Functional decomposition is a term that is used to describe a set of steps in which they break down the overall function of a device, system, or process into its smaller parts as per specifications by the producer. The decomposition of a process or function into smaller sub-functions can help project managers to determine how the individual functions or tasks help to achieve the overall project's goal. It mostly refers to the process of resolving a functional relationship into its constituent parts in such a way that the original function can be reconstructed from those parts by function composition. This is usually accomplished through thoughtful analysis and team discussions of project information. It helps us by providing a clear understanding and better specific design concept. The prime objective of this is to get a clear vision of all the functional requirements of the product and to search for further development of the design. The goal of function decomposition is to transform a complex task into a sequence of smaller tasks that are easier to understand, and then easier to find practical ways of achieving it.

Functional Decomposition Steps

There are four main steps of functional decomposition. These four steps are following-

Step 1: Find the basic functions

The first most important step is to find out the function that reflects the overall purpose of the design. There could be one or mostly two basic functions in every product design.

Step 2: Find the closest sub-functions

The second step is to analyze the sub-functions and list the sub-functions which are essential to get the success of the basic function.

Step 3: Assemble in the right order

The objective here is to understand which function comes first and what after that. It is a step-by step analysis of the main function of the product.

Step 4: Inspect the diagram

If there are functions that have been omitted, add them to the diagram.

5.2 Black Box

A black box is a system which can be viewed in terms of its inputs and outputs (or transfer characteristics), without any knowledge of its internal workings. There are generally three types of input and corresponding output in a black box. They are:

- Energy Flow
- Material Flow
- Information Flow

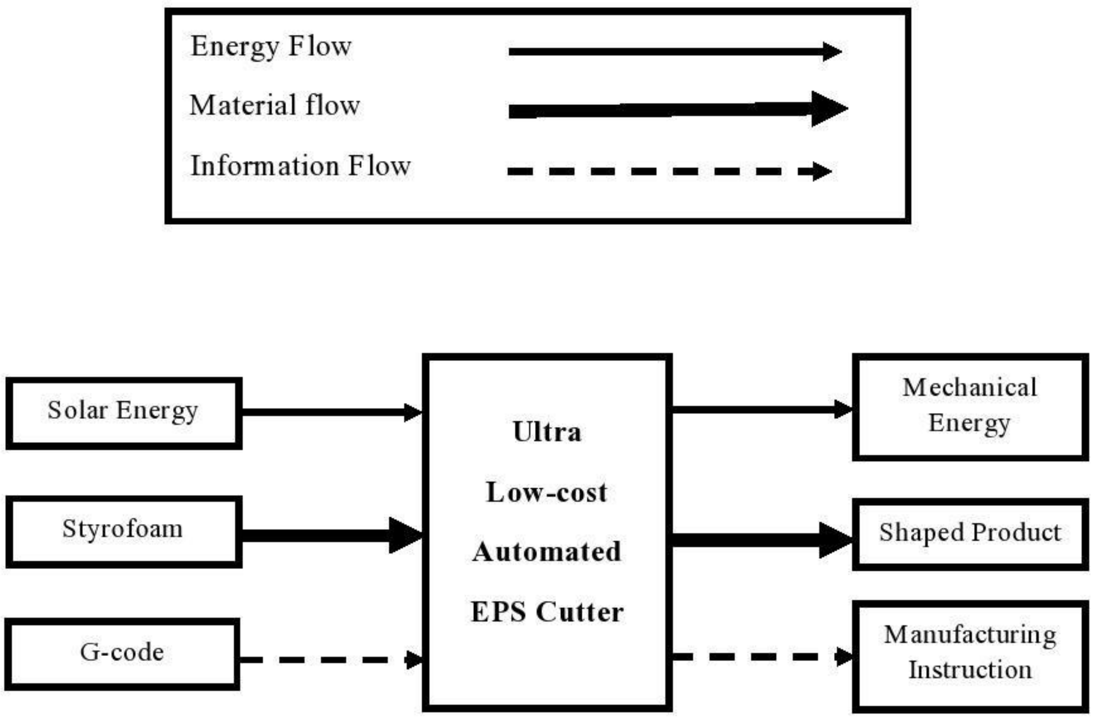


Figure: Black Box Diagram of **Ultra Low-cost Automated EPS Cutter**

5.3 Component Hierarchy

Hierarchy refers to a system that is assigned in the shape of a pyramid, with each row of objects directly beneath it. It divides the components into various major groups and subgroups to relate them.

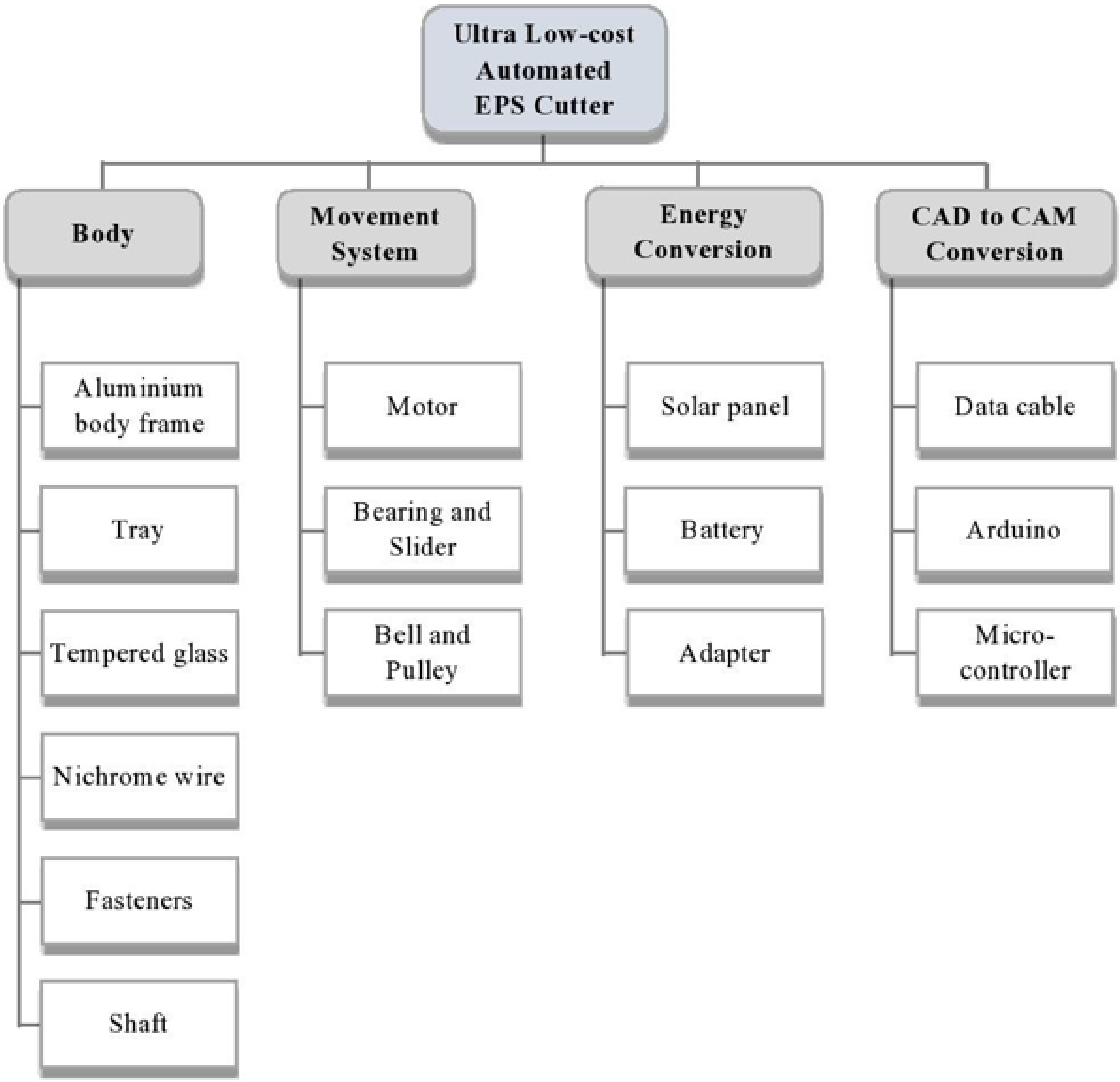


Figure: Component hierarchy of "**Ultra Low-Cost Automated EPS Cutter**"

5.4 Cluster Function Structure

Cluster function structure describes the processes, involved with individual features in a step-by-step manner. It helps to visualize the product functions and minimizes the possibility of any system error. Cluster function of our product, the **“Ultra Low-Cost Automated EPS Cutter”**

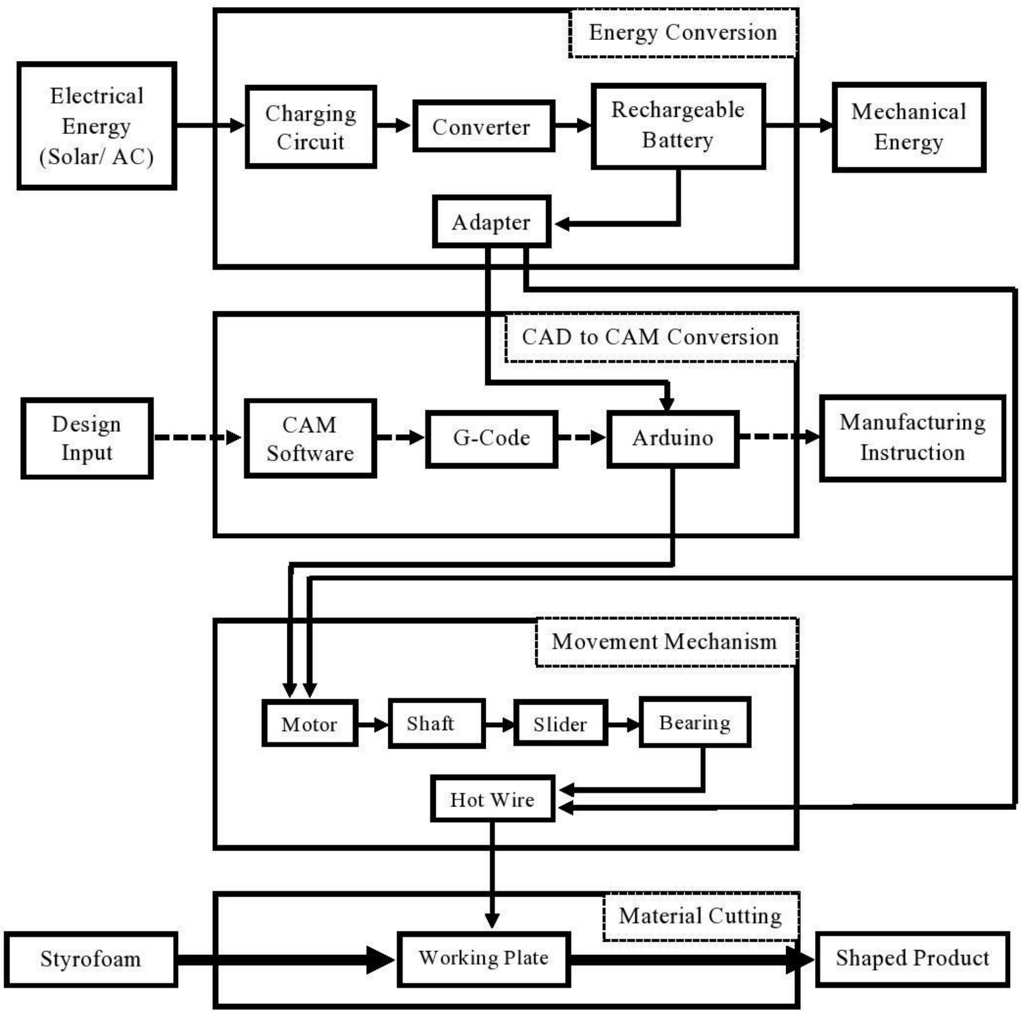


Figure: Cluster Function Structure of “**Ultra Low-Cost Automated EPS Cutter”**

5.5 Conclusion

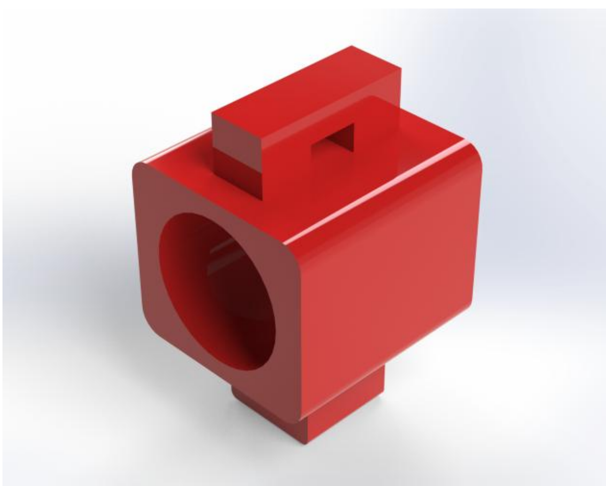
Decomposing our product by function, we break down a large, complex process into an array of smaller, simpler and better understanding processes. It provides specifications for both research and design. Also the black box model summarizes the energy, material and information flows input and output. We can get all the components and their working structure by component hierarchy. Finally, cluster function gives a visualization of the operation process and pathways. In this way, our product works.

Chapter 6
Design Analysis of Ultra Low-Cost Automated EPS Cutter

6.1 Parts of Ultra Low-Cost Automated EPS Cutter

Fig-1: T-slot Aluminum Profile

Fig-2: Linear Rails Rod



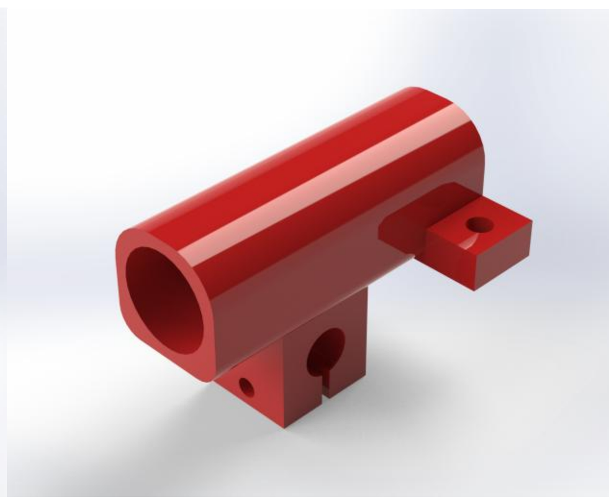


Fig-3: Sliding Block

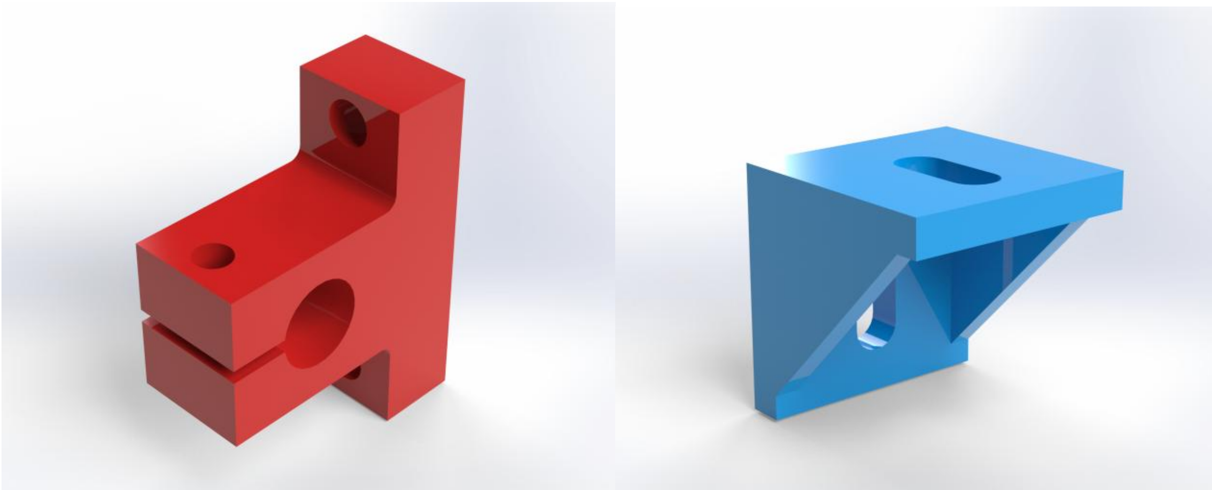


Fig-4: T-slot Profile Corner and Bracket

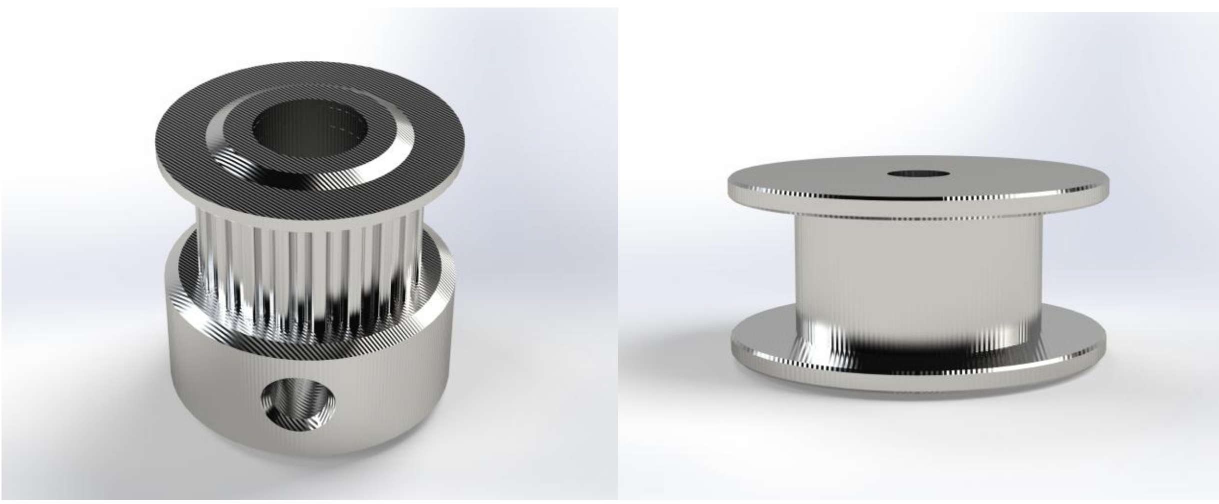


Fig-5: Tooth Pulley & Idler Pulley

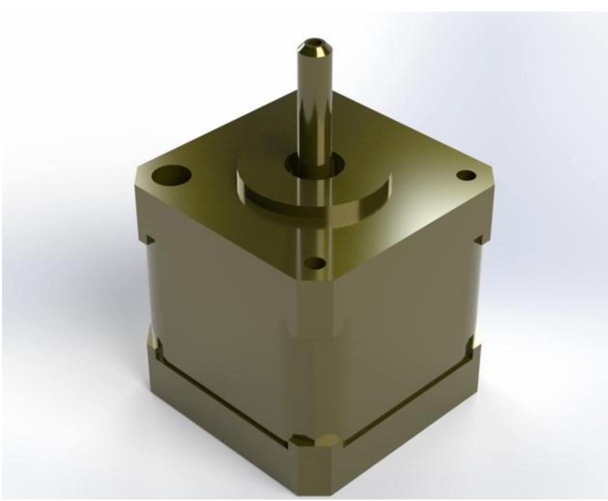


Fig-6: Stepper Motor

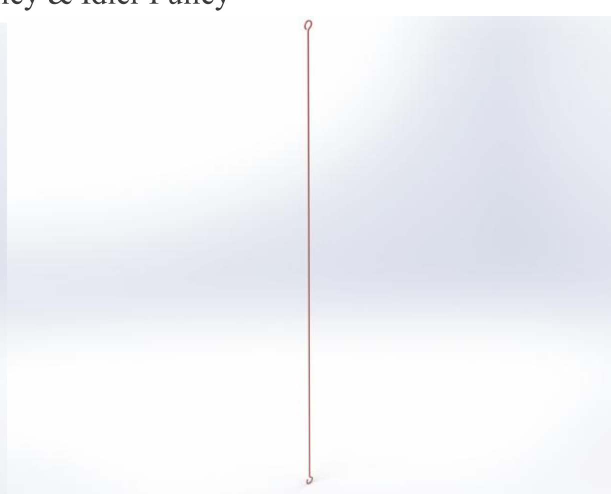


Fig-7: Hot wire

6.2 Final Assembly of Ultra Low-Cost Automated EPS Cutter

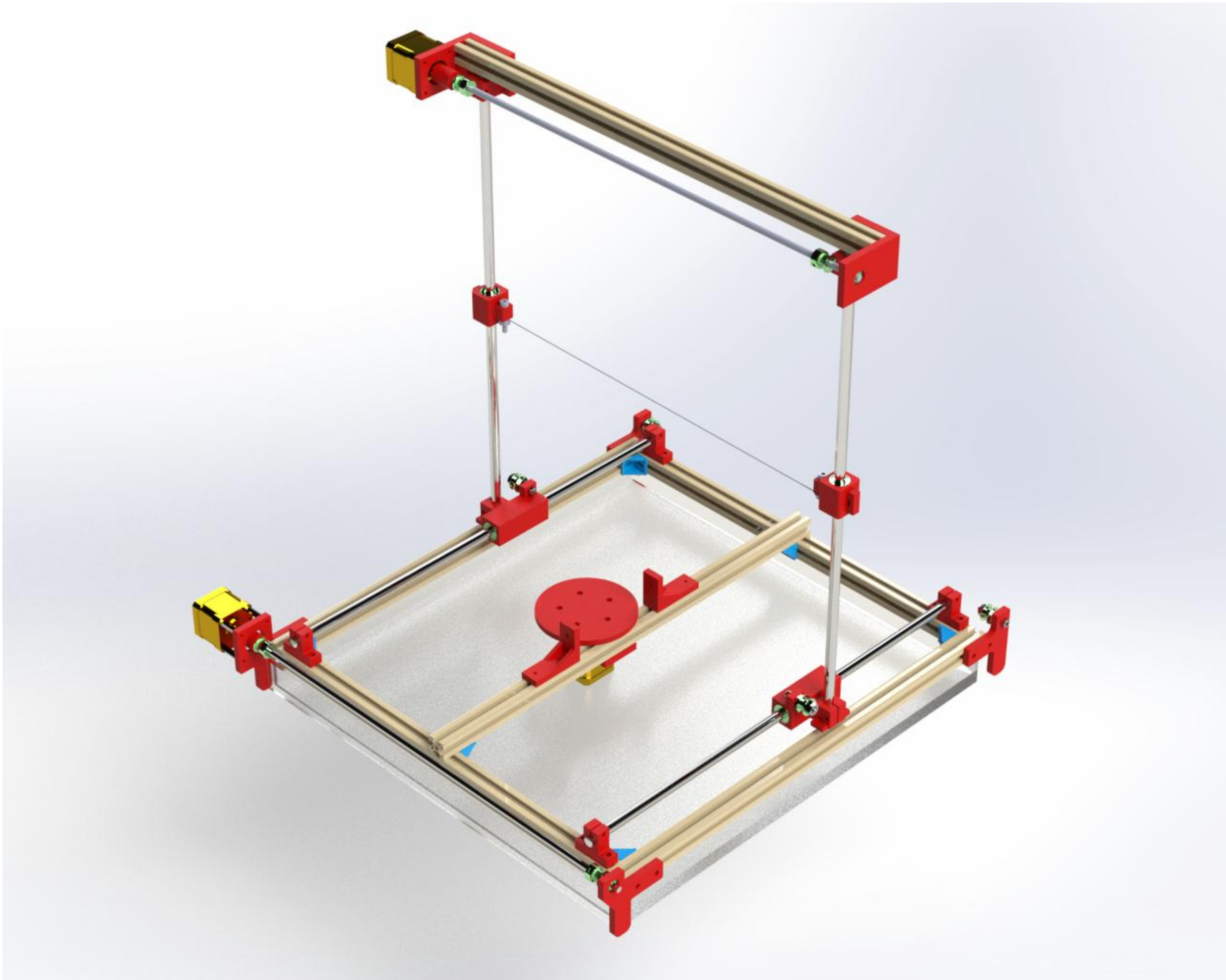


Fig-6: Ultra Low-Cost Automated EPS Cutter

Chapter 7
Selection of Alternate Material Using Weighted Average Method

7.1 Introduction

Material selection is an essential aspect of engineering processes of both products and production systems. The main goal of material selection is to minimize cost while meeting product performance goals. From the perspective of engineering design, it is a multiple attribute decision making problem. The selection of materials based on their limitations and possibilities in manufacturing is a complicated process. For making the selection process precise, logical, and simple, we are applying the weighted average method. In this process, we considered the objective weights of importance of the attributes and the subjective preferences to decide the integrated weights of importance of the attributes.

7.2 Structural parts of Ultra Low-Cost Automated EPS Cutter

- Body frame
- Hot-wire
- Solar panel
- Tempered glass
- Shaft
- Motor
- Bearing
- Belt and Pulley
- Tray

Above them, Body frame and Hot-wire materials should be selected by the Weighted Average Method.

7.3 Material Selection for Product Body Frame

7.3.1 Determination of Relative Importance of Goals Using Digital Logic Method

Selection Criteria	Number of positive decisions. N=n(n-1)/2=7(7-1)/2=21																					Positive Decisions	Relative Emphasis Co-Efficient, α
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21		
Cost	1	0	0	1	1	0																3	0.1428
Availability	0						0	0	0	1	1											2	0.095
Machinability		1					1					1	0	0	1							4	0.1904
Density			1					1				0				1	0	1				4	0.1904
Compressive Strength				0					1				1			0			1	1		4	0.1904
Young's Modulus					0					0				1			1		0		0	2	0.095
Hardness						1					0				0			0		0	1	2	0.095
Total Number of positive decisions																						21	Σα= 1

7.3.2 Calculation of the Performance Index

Selection Criteria	Weighting factor, α	Aluminum		Mild Steel	
		Scaled property, β (%)	Weighted score, αβ	Scaled property, β (%)	Weighted score, αβ
Cost	0.1428	66.67	9.52	100	14.28
Availability	0.095	100	9.5	100	9.5
Machinability	0.1904	100	19.04	75	14.28
Density	0.1904	100	19.04	34.52	6.57
Compressive Strength	0.1904	96.4	18.35	100	19.04
Young's Modulus	0.095	100	9.5	97.5	9.26
Hardness	0.095	100	9.5	34.5	1.2
Material Performance Index, y			94.45		73.63

7.3.3 Result

Material Performance Index of Aluminum (94.45) is greater than Mild Steel (73.63). So, we should select Aluminum for the body frame.

7.3.4 Material Property

Property	Aluminum	Mild Steel
Density (kg/m^3)	2710	7850
Brinell Hardness	240	120
Compressive Strength (MPa)	241	250
Young's Modulus (GPa)	69	200

7.3.5 Numerical Value (Rating) for Cost, Availability, Machinability

Very High	5
High	4
Medium	3
Low	2
Very Low	1

Selection Criteria	Aluminum	Mild Steel
Cost	3	2
Availability	4	4
Machinability	4	3

7.3.6 Formula Used

- **For goals:** Availability, Compressive Strength, Hardness, and Machinability.

$$Scaled\ property, \beta = \frac{Numerical\ Value\ of\ property}{Maximum\ value\ in\ list} \times 100\%$$

- **For goals:** Cost, Density and Young's Modulus.

$$Scaled\ property, \beta = \frac{Minimum\ value\ in\ list}{Numerical\ Value\ of\ property} \times 100\%$$

7.4 Material Selection for Cutting wire

7.4.1 Determination of Relative Importance of Goals Using Digital Logic Method

Selection Criteria	Number of positive decisions. N=n(n-1)/2=7(7-1)/2=21																					Positive Decisions	Relative Emphasis Co-Efficient, α
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21		
Cost	1	0	1	0	1	0																3	0.143
Availability	0						1	1	0	1	0											3	0.143
Melting Point		1					0					1	1	0	0							3	0.143
Density			0					0				0				0	0	1				1	0.048
Tensile Strength				1					1				0			1			0	0		3	0.143
Co-efficient of Thermal Expansion (m/˚C)					0					0				1			1		1		1	4	0.190
Sharpness						1					1				1			0		1	0	4	0.190
Total Number of positive decisions																						21	Σα= 1

7.4.2 Calculation of the Performance Index

Selection Criteria	Weighting factor, α	Tungsten		Nichrome Wire	
		Scaled property, β (%)	Weighted score, αβ	Scaled property, β (%)	Weighted score, αβ
Cost	0.143	50	7.15	100	14.3
Availability	0.143	50	7.15	100	14.3
Melting Point	0.143	100	14.3	41	5.863
Density	0.048	43.63	2.1	100	4.8
Tensile Strength	0.143	100	14.3	70.31	10.1
Co-efficient of Thermal Expansion (m/˚C)	0.190	100	19	32.85	6.24
Sharpness	0.190	50	9.5	100	19
Material Performance Index, y			73.5		74.6

7.4.3 Result

Material Performance Index of Nichrome Wire (74.6) is greater than Tungsten (73.5). So, we should select Nichrome Wire for the cutting wire.

7.4.4 Material Property

Property	Tungsten	Nichrome Wire
Density (kg/m^3)	19250	8400
Tensile Strength (MPa)	980	689
Co-efficient of Thermal Expansion (m/˚C)	4.6×10^{-6}	14×10^{-6}
Melting Point (°C)	3,422	1,400

7.4.5 Numerical Value (Rating) for Cost, Availability, Sharpness

Very High	5
High	4
Medium	3
Low	2
Very Low	1

Selection Criteria	Tungsten	Nichrome Wire
Cost	4	2
Availability	2	4
Sharpness	2	4

7.4.6 Formula Used

- **For goals:** Availability, Melting Point, Tensile Strength and Sharpness.

$$Scaled\ property, \beta = \frac{Numerical\ Value\ of\ property}{Maximum\ value\ in\ list} \times 100\%$$

- **For goals:** Cost, Density and Co-efficient of Thermal Expansion.

$$Scaled\ property, \beta = \frac{Minimum\ value\ in\ list}{Numerical\ Value\ of\ property} \times 100\%$$

7.5 Conclusion

By using weighted average method, we have selected the appropriate material for our product "**Ultra Low-Cost Automated EPS Cutter**" among different alternatives. Now we can easily make decision for our manufacturing materials without any hesitation. Thus, we can move to the next selection of the product design where we can choose the manufacturing criterion by implying weighted average method for different parts and process.

Chapter 8
Manufacturing Process Selection by Weighted Average Method

8.1 Introduction

Manufacturing process selection refers to deciding on the way production of goods or services will be organized. The term manufacturing processes used here represents the main shape-generating methods such as shearing and bending, tube drawing, etc. processes, as well as traditional and nontraditional machining processes. The first step in manufacturing process selection is to establish selection criteria based on key process selection drivers: manufacturing volumes, value of the product, part geometry, required tolerances, and required material. The main goal of selection is to identify systematically the match between the requirements of the design and the capabilities of processes. In most cases there are several processes that can be used for a component and final selection depends on a large number of factors.

8.2 Manufacturing Process required for Ultra Low-cost Automated EPS Cutter

- We would like to purchase some raw materials like Aluminum.
- Desired size and shape will be achieved through further manufacturing process.
- We need to cut Aluminum and use nut-bolt. This will make up the whole frame of the body.
- For permanent joint we are going to use arc welding and for temporary joint we are going to use nut-bolt. Temporary joints are basically for flexible and adjustable parts.
- Some parts of our product are needed to be purchased. Selection processes for those are not included here.

8.3 Manufacturing Process for Body Frame

8.3.1 Determination of Relative Importance of Goals Using Digital Logic Method

Selection Criteria	Number of Positive Decisions, N=n(n-1)/2=5(5-1)/2=10										Positive Decisions	Relative Emphasis Coefficient, α
	1	2	3	4	5	6	7	8	9	10		
Cost	1	1	0	1							3	0.3
Power Requirement	0				0	0	0				0	0.0
Ease of Use		0			1			1	0		2	0.2
Dimensional Accuracy			1			1		0		0	2	0.2
Surface Smoothness				0			1		1	1	3	0.3
Total Number of Positive Decisions											10	**$\sum \alpha$ =1.00**

8.3.2 Calculation of the Performance Index

Selection Criteria	Weighting factor, α	Shearing and Bending		Hand Cutting and Hammering	
		Scaled property, β (%)	Weighted score, αβ	Scaled property, β (%)	Weighted score, αβ
Cost	0.3	50	15	100	30
Power Requirement	0.0	100	0	66.67	0
Ease of Use	0.2	100	20	50	10
Dimensional Accuracy	0.2	100	20	75	15
Surface Smoothness	0.3	100	30	60	18
Process Performance Index, γ			**85**		73

8.3.3 Result

Process Performance Index is the greatest (85) for Shearing and Bending process. So, we should select this process for cutting of Aluminum for frame and container.

8.3.4 Chart Containing Relative Evaluations of Specifications

Very High	5
High	4
Medium	3
Low	2
Very Low	1

8.3.5 Chart Containing Assigned Values for Specifications

Selection Criteria	**Shearing and Bending**	**Hand Cutting and Hammering**
Cost	4	2
Power Requirement	2	3
Ease of Use	4	2
Dimensional Accuracy	4	3
Surface Smoothness	5	3

8.3.6 Formula Used

- **For goals:** Ease of use, Dimensional Accuracy, Surface Smoothness –

$$\text{Scaled Property, } \beta = \frac{Numerical\ value\ of\ property}{Maximum\ value\ in\ list} \times 100\%$$

- **For goals:** Cost, Power Requirement –

$$\text{Scaled Property, } \beta = \frac{Minimum\ value\ in\ list}{Numerical\ value\ of\ property} \times 100\%$$

8.4 Joining Process for Different Parts

8.4.a Temporary Joining

8.4.a.1 Determination of Relative Importance of Goals Using Digital Logic Method

Selection Criteria	**Number of Positive Decisions, N=n(n-1)/2=5(5-1)/2=10**										**Positive Decisions**	**Relative Emphasis Coefficient, α**
	1	2	3	4	5	6	7	8	9	10		
Cost	0	1	0	1							2	0.20
Strength	1				1	1	1				4	0.40
Availability		0			0			0	1		1	0.10
Force Requirement			1			0		1		0	2	0.20
Flexibility				0			0		0	1	1	0.10
Total Number of Positive Decisions											10	**$\sum \alpha$ =1.00**

8.4.a.2 Calculation of the Performance Index

Selection Criteria	**Weighting factor, α**	**Latch**		**Nut-Bolt**	
		Scaled property, β (%)	**Weighted score, αβ**	**Scaled property, β (%)**	**Weighted score, αβ**
Cost	0.20	100	20	75	15
Strength	0.40	50	20	100	40
Availability	0.10	75	7.5	100	10
Force Requirement	0.20	100	20	75	15
Flexibility	0.10	75	7.5	100	10
Process Performance Index, γ			75		**90**

8.4.a.3 Result

Process Performance Index is greater for Nut-Bolt (90) than Latch (75). So, we going to select Nut-Bolt for temporary joining process.

8.4.a.4 Chart Containing Relative Evaluations of Specifications

Very High	5
High	4
Medium	3
Low	2
Very Low	1

8.4.a.5 Chart Containing Assigned Values for Specifications

Selection Criteria	Latch	Nut-Bolt
Cost	3	4
Strength	2	3
Availability	3	4
Force Requirement	4	3
Flexibility	3	4

8.4.a.6 Formula Used

- **For goals:** Strength, Availability, Force Requirement, Flexibility –

$$\text{Scaled Property, } \beta = \frac{Numerical\ value\ of\ property}{Maximum\ value\ in\ list} \times 100\%$$

- **For goals:** Cost –

$$\text{Scaled Property, } \beta = \frac{Minimum\ value\ in\ list}{Numerical\ value\ of\ property} \times 100\%$$

8.4.b Permanent Joining

8.4.b.1 Determination of Relative Importance of Goals Using Digital Logic Method

Selection Criteria	**Number of Positive Decisions, N=n(n-1)/2=6(6-1)/2=15**															**Positive Decisions**	**Relative Emphasis Coefficient, α**
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15		
Cost	1	1	0	0	1											3	0.2
Availability	0					1	0	0	1							2	0.133
Ease of Operation		0				0				0	1	1				2	0.133
Strength			1				1			1			0	1		4	0.267
Dimensional Accuracy				1				1			0		1		0	3	0.2
Power Requirement					0				0			0		0	1	1	0.067
Total Number of Positive Decisions																15	**$\sum \alpha = 1.00$**

8.4.b.2 Calculation of the Performance Index

Selection Criteria	**Weighting factor, α**	**Gas Welding**		**Arc Welding**	
		Scaled property, β (%)	**Weighted score, αβ**	**Scaled property, β (%)**	**Weighted score, αβ**
Cost	0.2	100	20	75	15
Availability	0.133	100	13.3	100	13.3
Ease of Operation	0.133	60	7.98	100	13.3
Strength	0.267	60	16.02	100	26.7
Dimensional Accuracy	0.2	50	10	100	20
Power Requirement	0.067	100	6.7	75	5.025
Process Performance Index, γ			74		**80.025**

8.4.b.3 Result

Process Performance Index is the greatest (**80.025**) for Arc Welding. So, we should select this for permanent joining process.

8.4.b.4 Chart Containing Relative Evaluations of Specifications

Very High	5
High	4
Medium	3
Low	2
Very Low	1

8.4.b.5 Chart Containing Assigned Values for Specifications

Selection Criteria	Gas Welding	Arc Welding
Cost	3	4
Availability	4	4
Ease of Operation	3	5
Strength	3	5
Dimensional Accuracy	2	4
Power Requirement	3	4

8.4.b.6 Formula Used

- **For goals:** Availability, Ease of Operation, Strength, Dimensional Accuracy, Power Requirement –

$$\text{Scaled Property, } \beta = \frac{Numerical\ value\ of\ property}{Maximum\ value\ in\ list} \times 100\%$$

- **For goals:** Cost –

$$\text{Scaled Property, } \beta = \frac{Minimum\ value\ in\ list}{Numerical\ value\ of\ property} \times 100\%$$

8.5 Finishing Process

8.5.1 Determination of Relative Importance of Goals Using Digital Logic Method

Selection Criteria	**Number of Positive Decisions, N=n(n-1)/2=5(5-1)/2=10**										**Positive Decisions**	**Relative Emphasis Coefficient, α**
	1	2	3	4	5	6	7	8	9	10		
Cost	1	0	1	1							3	0.3
Availability	0				1	1	1				3	0.3
Material Wastage		1			0			0	1		2	0.2
Durability			0			0		1		0	1	0.1
Smoothness				0			0		0	1	1	0.1
Total Number of Positive Decisions											10	**∑ α =1.00**

8.5.2 Calculation of the Performance Index

Selection Criteria	**Weighting factor, α**	**Non-Precision Grinding**		**Honing**	
		Scaled property, β (%)	**Weighted score, αβ**	**Scaled property, β (%)**	**Weighted score, αβ**
Cost	0.3	100	30	80	24
Availability	0.3	100	30	80	24
Material Wastage	0.2	100	20	60	12
Durability	0.1	100	10	60	6
Smoothness	0.1	60	6	100	10
Process Performance Index, γ			**96**		76

8.5.3 Result

Process Performance Index is the greatest (96) for Non-Precision Grinding. So, we should select this for finishing process.

8.5.4 Chart Containing Relative Evaluations of Specifications

Very High	5
High	4
Medium	3
Low	2
Very Low	1

8.5.5 Chart Containing Assigned Values for Specifications

Selection Criteria	**Non-Precision Grinding**	**Honing**
Cost	4	5
Availability	5	4
Material Wastage	3	5
Durability	5	5
Smoothness	3	5

8.5.6 Formula Used

- **For goals:** Smoothness, Availability, Durability –

$$\text{Scaled Property, } \beta = \frac{\textit{Numerical value of property}}{\textit{Maximum value in list}} \times 100\%$$

- **For goals:** Cost, Material Wastage –

$$\text{Scaled Property, } \beta = \frac{\textit{Minimum value in list}}{\textit{Numerical value of property}} \times 100\%$$

8.6 Coloring Process

8.6.1 Determination of Relative Importance of Goals Using Digital Logic Method

Selection Criteria	**Number of Positive Decisions, N=n(n-1)/2=6(6-1)/2=15**															**Positive Decisions**	**Relative Emphasis Coefficien t, α**
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15		
Cost	1	0	1	1	0											3	0.2
Availability	0					0	0	0	0							0	0.0
Ease of Operation		1				1				1	0	0				3	0.2
Durability			0				1			0			0	1		2	0.133
Wastage of Color				0				1			1		1		0	3	0.2
Smoothness					1				1			1		0	1	4	0.267
Total Number of Positive Decisions																15	**$\sum$ α = 1.00**

8.6.2 Calculation of the Performance Index

Selection Criteria	**Weighting factor, α**	**Brush Painting**		**Color Spraying**	
		Scaled property, β (%)	**Weighted score, αβ**	**Scaled property, β (%)**	**Weighted score, αβ**
Cost	0.2	100	20	60	12
Availability	0.0	100	0	80	0
Ease of Operation	0.2	80	16	100	20
Durability	0.133	80	10.64	100	13.3
Wastage of Color	0.2	100	20	60	12
Smoothness	0.267	100	26.7	80	21.36
Process Performance Index, γ			**93.34**		78.66

8.6.3 Result

Process Performance Index is the greatest (93.34) for Brush Painting. So, we should select this for coloring process.

8.6.4 Chart Containing Relative Evaluations of Specifications

Very High	5
High	4
Medium	3
Low	2
Very Low	1

8.6.5 Chart Containing Assigned Values for Specifications

Selection Criteria	Brush Painting	Color Spraying
Cost	3	5
Availability	5	4
Ease of Operation	4	5
Durability	4	5
Wastage of Color	3	5
Smoothness	5	4

8.6.6 Formula Used

- **For goals:** Availability, Ease of Operation, Durability, Smoothness –

$$\text{Scaled Property, } \beta = \frac{Numerical\ value\ of\ property}{Maximum\ value\ in\ list} \times 100\%$$

- **For goals:** Cost, Wastage of Color –

$$\text{Scaled Property, } \beta = \frac{Minimum\ value\ in\ list}{Numerical\ value\ of\ property} \times 100\%$$

8.7 Conclusion

We conclude that we have selected manufacturing process, joining process, finishing process and coloring process for our ‘**Ultra Low-cost Automated EPS Cutter**’ from a variety of alternatives by means of systematic procedure of Weighted Average Method. Now we would like to implement these processes to fulfill our design.

Chapter 9
Cost Analysis

9.1 Introduction

In business economics functional cost analysis is a method that can be applied to examine the component costs of a product in relation to the value as perceived by the customer. It is the process of tracking and analyzing all of the costs incurred in the production and sale of a product or service, including direct costs and overhead costs. The analysis gives clarity to unpredictable situations. The process is quite different for prototype making and actual mass manufacturing. In mass production the only cost that is to be considered is manufacturing cost or product cost. So, cost analysis is done for the actual mass manufacturing only. The outcome of the analysis is to improve the value of the product while maintaining costs or reduce the costs of the product without reducing value.

Cost analysis is a very important part of product design which provides data for overall raw material cost, direct labor cost of machineries, other purchasing costs and administrative cost. Cost analysis is a very important part of product design which provides data for overall raw material cost, direct labor cost of machineries, other purchasing costs and administrative cost. The goal with this thesis is to help, evaluate and support improvements within the product development projects when it comes to product cost calculation.

In this report we show the cost analysis for our **Ultra Low-Cost Automated EPS Cutter**. We also determine the BEP (Breakeven point) to check whether this project is feasible and easy to make.

We should consider some factors while doing cost analysis. Those are as follows-

- Present market condition
- Pre-production
- Competitive pricing
- Customer’s affordability
- Long term planning
- Capacity requirement
- Allocation of cost
- Feasibility and motivation
- Manufacturing cost
- Administrative expenses
- Selling expense
- Furniture and computer expense

9.2 Machine and Associated Cost

1. Aluminum Extrusion Plant (V-Fast Furnaces & Technologies, India)
Buying Cost: Tk. 1200000
Life: 20 years
Salvage: Tk. 150000
Quantity: 1
Considering salvage value, buying cost = 1,200,000 – 150,000(P|F, 10%, 20)
= **1,177,710** Tk.

2. Semi-Automatic Turning Machine (Atul Machine Tools, India)
Buying Cost: Tk. 160,000
Life: 20 years
Salvage: Tk. 20,000
Quantity: 1
Considering salvage value, buying cost = 160000 - 20000 (P|F, 10%, 20)
= **157,028** Tk.

3. Plastic Injection
Buying Cost: Tk. 600,000
Life: 20 years
Salvage: Tk. 60,000
Quantity: 1
Considering salvage value, buying cost = 600,000 – 60,000 (P|F, 10%, 20)
= **591,084** Tk.

4. Gear Hobbing Machine (Hebei Dechuang Chemical Equipment Co., Ltd, China, Model No: Y3150 3180)
Buying Cost: Tk. 300,000
Life: 20 years
Salvage: Tk. 30,000
Quantity: 1
Considering salvage value, buying cost = 300,000 – 30,000 (P|F, 10%, 20)
= **295,542** Tk.

5. PVC Belt making machine (Sant Engineering Industries, China, Model No: Y3150 3180)
Buying Cost: Tk. 1,200,000
Life: 20 years
Salvage: Tk. 120,000
Quantity: 1
Considering salvage value, buying cost = 1,200,000 - 120,000 (P|F, 10%, 20)
= **1,182,168** Tk.

Total Machine Cost = **3,403,532 Tk**

9.3 Cost of Furniture, Computer & Other Accessories

• Total furniture & accessories cost for office: Tk. 100,000
Life: 10 years
Salvage: Tk. 25,000
Considering salvage value, buying cost = 100,000 – 25,000 (P|F, 10%, 10)
= **90,362.5 Tk.**

• Computer cost: Tk. 55,000
Quantity: 3
Life: 10 years
Salvage: Tk. 30,000
Considering salvage value, buying cost = 165,000 –60,000 (P|F, 10%, 10)
= 141,870 Tk.

Total other accessories cost = **90,362.5 + 141,870**= **232,232.5** Tk.

9.4 Costs of Raw Materials (per unit of product)
1. **Aluminum**
Sheet:
Raw material required = 10kg
Market price of raw material = 80 tk/kg.
Total Raw Material cost = 80 Tk.

2. Brush Paint
Raw material (Color) required = 1-liter
Market price of raw material = 400tk/liter
Total raw material cost = 400 Tk.

3. **Rubber**
Raw material required = 100 gm
Market price of raw material = 3 tk/kg
Total raw material cost = 3 Tk

4. **Iron**
Raw material required = 300 gm
Market price of raw material = 70 tk/kg
Total raw material cost = 21 Tk

5. **Plastic**
Raw material required = 100 gm
Market price of raw material = 50 tk/kg
Total raw material cost = 5 Tk

9.5 Manufacturing Costs of Different Operations (Per Month)

- **Measuring**
  Labor cost
  No of workers = 2
  Wage/labor = 10,000 Tk.
  Total Labor cost = 20,000 Tk.
- **Cutting**
  Labor cost
  No of workers = 2
  Wage/labor = 9,000 Tk.
  Total labor cost = 18,000Tk.

- **Welding**
  Labor cost
  No of workers = 2
  Wage/labor = 8, 000 Tk.
  Total Labor cost = 16,000 Tk.

- **Assembly and Joining Operation**
  Labor cost
  No of workers = 2
  Wage/labor = 14,500 Tk.
  Total Labor cost = 29,000 Tk.

- **Finishing Operation**
  Labor cost
  No of workers = 2
  Wage/labor = 9,000 Tk.
  Total Labor cost = 18,000 Tk.

- **Coloring Operation**
  Labor cost
  No of workers = 2
  Wage/labor = 9, 000 Tk.
  Total Labor cost = 18,000 Tk.

- **Packaging Operation**
  Labor cost
  No of workers = 2
  Wage/labor = 8,000 Tk.
  Total Labor cost = 16,000 Tk

Total raw material cost: **509 Tk. Per unit**
Total labor cost: **135,000 Tk. Per month**
Total unit of production per year: **1,800**

Total raw material cost per year: (1,800 × 509) = 916,200 Tk.
Total labor cost per year: 135,000 × 12 = 1,620,000Tk.

9.6 Purchasing Cost

The list of parts which we are going to be purchased is as follows:

Name of the Parts	Required Quantity	Unit Price in Tk.	Price in Tk.
Nut-Bolt	Handful	-	300
T-slot profile corner	6 pc.	10	60
Tempered Glass	10 sqr feet	150	1,500
Bearing	8 pc.	100	800
Motor	3 pc.	300	900
Solar Panel	1 pc.	2,000	2000
Battery	1 pc.	1,500	1,500
Arduino	1 pc.	800	800
Hot wire	1 pc.	40	40
Total Cost			7,900

Quantity discount is considered here as some of the above listed parts will be purchased in bundle rather than separately. According to the present market condition we estimate that about **25% discount** will be available if we buy these parts in large amount.

So, total purchasing cost for parts = 7,900 × (1-25%) = 5,925Tk.

Total purchasing cost per year = 5,925 × 1,800 = **10,665,000Tk.**

9.7 Manufacturing Cost

Direct Material Cost
Raw Material Cost: **916,200** Tk.
Purchasing Cost: **10,665,000** Tk.
Direct material used in production per year: (**916,200** + **10,665,000**) = **11,581,200** Tk.

Direct Labor Cost
Labor Cost = **1,620,000**Tk.

Manufacturing Overhead Cost per Month

Post	No. of Post	Salary/Person	Total (Tk.)
Production Manager	1	45,000	45,000
Manufacturing Engineer	1	40,000	40,000
Assembly Manager	1	30,000	30,000
Quality Control Engineer	1	30,000	30,000
Supply Chain Manager	1	30,000	30,000
Factory Rent	-	40,000	50,000
Utilities and Maintenance	-	-	40,000
Total			265,000

Total manufacturing overhead per year: 265000 × 12 = **3,180,000 Tk**.
Total manufacturing cost per year: **11,581,200** + **1,620,000+ 3180000**
= 16,381,200 Tk.

9.8 Administrative and Selling Cost

9.8.1. Administrative Cost

Post	No. of Post	Salary/Person	Total (Tk.)
Chief Executive Officer	1	70,000	70,000
General Manager	1	45,000	45,000
HR Manager	1	30,000	30,000
Accountant	1	30,000	30,000
Clerk	2	10,000	10,000
Guard	2	8,000	8,000
Power Bill	-	60,000	60,000
Water Bill	-	2,000	2,000
Office Rent	-	20,000	20,000
Total			275,000

9.8.2. Selling Expenses

Cost Item	No. of Post	Salary/Person	Total (Tk.)
Marketing Executive	1	20,000	20,000
Advertising		25,000	25,000
Total			45,000

Total Selling and Administrative expenses per month: (275,000 + 45,000) = **320,000** Tk.
Total Selling and Administrative expenses per year: (320,000 × 12) = **3,840,000 Tk.**
Total unit of production per year: **1,800**

9.9 Break Even Analysis

Fixed Cost

- Machine Cost = **3,403,532** Tk.
- Furniture, Computer & other Accessories Cost = **232,232.5** Tk.
- Labor Cost = **1,620,000Tk**.
- Fixed Manufacturing Overhead Cost = **3,180,000** Tk.
- Fixed Selling and Administrative Cost = **3,840,000** Tk.

Total amount of Fixed Cost = **12,275,764.5** Tk Per year

Variable Cost (for the 1,800 products in the 1st year)

- Raw Material Cost = **916,200**Tk.
- Purchasing Cost per year = **10,665,000** Tk.

Total Variable Cost per year = **11,581,200** Tk.
Total variable cost per unit production = **11,581,200** ÷ **18,00** = **6,43**4 Tk.
Selling Price per unit = **6,43**4 + 60 % of **6,43**4 (expecting 60 % profit on sale)
= **10,295 Tk.**

The Equation for Break Even Analysis
At break-even point,

$$\boldsymbol{Selling\ Price \times unit\ of\ production, (x) = Fixed\ Cost + Variable\ Cost}$$

Total Amount of Fixed Cost = **12,275,764.5**Tk.
The total variable cost per unit = **6,434**Tk.
Selling Price per unit = **10,295** Tk.

So, the equation stands as,
10,295 *X = **12,275,764.5**+ **6,434 *X**
this yields **X** = 3,179.4 ≈ **3,180 units**

So, our payback period is almost for approximately **1 year 9 months.**

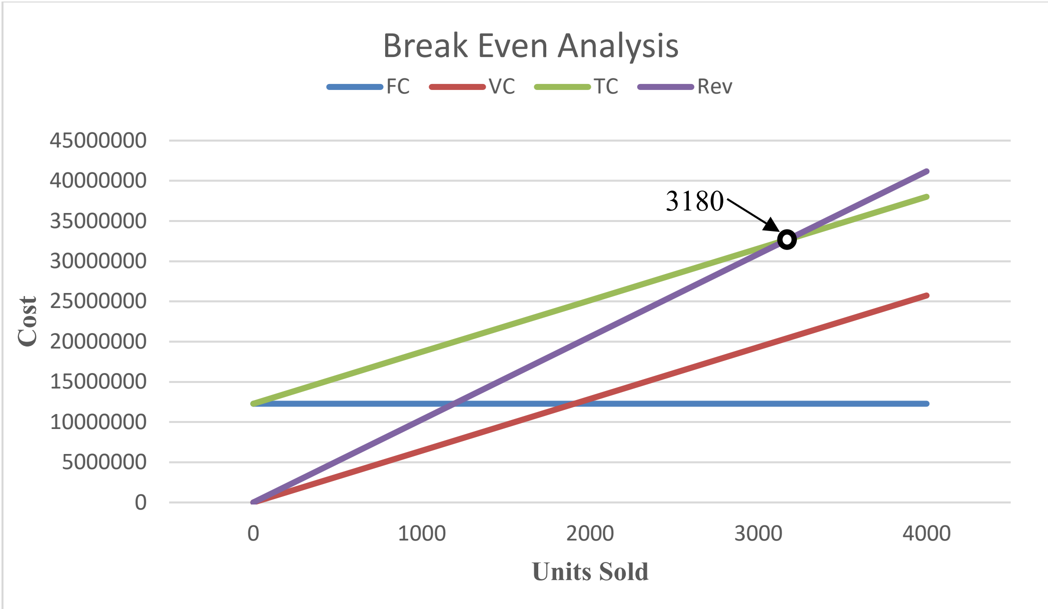


Figure: Break Even Analysis

9.10 Conclusion

The cost of the materials and machines are assumed according to our field survey to various markets. Thus, it is basically an estimation to determine the cost analysis of the product which is mostly near to the real cost. Our break-even point is 3,180 units. It will take 1 year and 9 months to manufacture. Thus, after this time we will be able to defend the loss and start to profit. So, hope that our product will be a profitable one indeed.

Chapter 10
Conclusion

Expanded Polystyrene (EPS) is very popular for its unique materials properties. It is rigid, tough and light weight thermoplastic product. Besides, it can withstand high compressive load and back fill force. Also, its closed cell structure provides minimal water absorption and low vapor permanence.

For its uniqueness, it is widely used through the world. It is used in building constructions, food packaging, industrial packaging, sports helmets, infant car seats, chairs, arts and crafts, thermal vessels, aircraft applications and so on.

Unfortunately, in our country, this EPS is cutting manually or semi-automatically. Thus, the workers have to go through risk of cutting finger and harsh noise. Moreover, a lot of cellular EPS creates place dirty. Also, it not only takes a lot of time for cutting operation but also provides rough surface finish.

To make our country self-dependent in EPS cutting operations by avoiding the import of the EPS cutter we designed “Ultra Low-Cost EPS Cutter”. It will also solve the existing cutting problems in a low budget. From surveying, we understand that our market is in lack of such a product. Thus, we expect that our product will flourish the market by meeting our customer’s expectations.

References

[1] Sulkowski W W Nowak K Sulkowska A Wolinska A Bajdur W M Pentak D Mikula B 2009 Study of the Sulfonation of Expanded Polystyrene Waste and of Properties of the Products Obtained Pure Appl.Chem 81 2417-2424

[2]Panchal, H.B., Mayur, S., Patel, P.D. and Padia, U.N., Arduino Based CNC Machine. International Journal of Research in Science & Engineering, P-ISSN, pp.2394-8280.

[3]Panchal, H.B., Mayur, S., Patel, P.D. and Padia, U.N., Arduino Based CNC Machine. International Journal of Research in Science & Engineering, P-ISSN, pp.2394-8280.

[4]Aitchison, D.R., Brooks, H.L., Bain, J.D. and Pons, D., 2011. Rapid manufacturing facilitation through optimal machining prediction of polystyrene foam: The findings reported in this paper provide a model for consistently achieving optimal cutting conditions when shaping foam plastic, such as polystyrene, with an electrically heated wire-based cutting tool. Virtual and Physical Prototyping, 6(1), pp.41-46

[5]Mamadjanov, A.M., Yusupov, S.M. and Sadirov, S., 2021. Advantages and the future of cnc machines. Scientific progress, 2(1), pp.1638-1647.

[6]H. Brooks, "Plastic Foam Cutting Mechanics for Rapid Prototyping and Manufacturing Purposes" (Ph.DThesis), Mechanical Engineeringat the University of Canterbury , Christ church , New Zealand , 2009

- Shigley's Mechanical Engineering Design, 10th Edition by Richard G. Budynas & J. Keith Nisbett
- Manufacturing Engineering & Technology, 6th Edition by Serope Kalpakjian & Steven Schmid
- Engineering Metrology by R. K. Jain & Khanna Publisher
- A Textbook of Machine Design by RS Khurmi & JK Gupta
- Engineering Economy by Leland Blank & Anthony Tarquin

Appendix

Appendix A: Questionnaire for the Survey of Ultra Low-Cost Automated EPS Cutter

What is Your Name?

1. What type of cutter do you use while cutting foam?
   a) Manual
   b) Semi-Automatic
   c) Automatic
   d) Others

2. Could you cut your desired shape by using a knife?
   a) Yes
   b) No
   c) Maybe

3. What makes you uncomfortable during cutting foam manually? (You can choose multiple option)
   a) Rough and irregular surface
   b) Slipping of cutter while cutting
   c) None

4. What issue makes you irritated when you try to cut a foam?(You can choose multiple option)
   a) Noise
   b) Dust
   c) Vibration during cutting

5. What do you think about the accuracy of manual foam cutting?
   a) Low
   b) Moderate
   c) High

6. What is the level of foam cutting skill of you or your organization?
   a) Low
   b) Moderate
   c) High

7. How much average time is taken by manual foam cutter, if you cut 'A' letter in regular shape?
   a) Below 15 Minutes
   b) 15 Minutes - 30 Minutes
   c) Above 30 Minutes

8. Are you or your organization familiar with designing software?
    a) Yes
    b) No
    c) Maybe

9. If an automated foam cutter is made, would you prefer to buy it?
    a) Yes
    b) No
    c) Maybe

10. What should be your expected price? (You can choose multiple option)
    a) 15k - 18k Taka
    b) 18k - 22k Taka
    c) 22k - 25k Taka
    d) More than 25k Taka

11. What do you want from this machine?
    a) To get product in less time
    b) To get a product in less cost
    c) Both

12. What should be the lifetime of this product in your opinion?
    a) Below 5 Years
    b) 5 Years - 10 Years
    c) Above 10 Years

13. What should be most likely to be reduced while using this product in your opinion?(You can choose multiple option)
    a) Time
    b) Cost
    c) Accidents

14. Would you like to recommend others to buy this product?
    a) If their work is related
    b) Maybe
    c) If I feel everything is reasonable.

15. What do you suggest to modify this product?

    ……………………………………………………………..
    ……………………………………………………………..

Appendix B: Properties of Metals and their alloy, Plastics

These following tables were used during material selection using a weighted average method.

Table 1. Material properties of aluminum alloy 6061-T6, alumina and steel.

Material	Temperature (°C)	Thermal conductivity ($W.m^{-1}.K^{-1}$)	Heat capacity ($J.kg^{-1}.K^{-1}$)	Density ($kg.m^{-3}$)	Thermal expansion ($\times 10^{-6}\ K^{-1}$)	Young's modulus (GPa)	Yield stress (MPa)	Poisson's Ratio	Melting point (°C)
Aluminum alloy 6061-T6	0	162	917	2703	22.4	69.7	277.7	0.33	582-652
	98	177	978	2685	24.61	66.2	264.6		
	201	192	1028	2657	26.6	59.2	218.6		
	316	207	1078	2630	27.6	47.78	66.2		
	428	223	1133	2602	29.6	31.72	17.9		
	571	253	1230	2574	34.2	0	0		
Alumina	25-727	16.7	950	3950	7.5	300	-	0.2	2127
Steel	0	34.5	470	7800	10.1-16.6	68.9-207	-	0.23-0.3	1230-1530
	98	345	485						
	201	33.8	520						
	316	31	560						
	428	28.5	620						
	571	26.8	700						
	650	25.8	760						

MATERIAL	Type	Cost ($/kg)	Density (ρ,Mg/m^3)	Young's Modulus (E, GPa)	Shear Modulus (G, GPa)	Poisson's Ratio (ν)	Yield Stress (σ_Y, MPa)	UTS (σ_f,MPa)	Breaking strain (ε_f, %)	Fracture Toughness (K_c,MN m$^{-3/2}$)	Thermal Expansion (α,10^{-6}/C)
Alumina (Al_2O_3)	ceramic	1.90	3.9	390	125	0.26	4800	35	0.0	4.4	8.1
Aluminum alloy (7075-T6)	metal	1.80	2.7	70	28	0.34	500	570	12	28	33
Beryllium alloy	metal	315.00	2.9	245	110	0.12	360	500	6.0	5.0	14
Bone (compact)	natural	1.90	2.0	14	3.5	0.43	100	100	9.0	5.0	20
Brass (70Cu30Zn, annealed)	metal	2.20	8.4	130	39	0.33	75	325	70.0	80	20
Cermets (Co/WC)	composite	78.60	11.5	470	200	0.30	650	1200	2.5	13	5.8
CFRP Laminate (graphite)	composite	110.00	1.5	1.5	53	0.28	200	550	2.0	38	12
Concrete	ceramic	0.05	2.5	48	20	0.20	25	3.0	0.0	0.75	11
Copper alloys	metal	2.25	8.3	135	50	0.35	510	720	0.3	94	18
Cork	natural	9.95	0.18	0.032	0.005	0.25	1.4	1.5	80	0.074	180
Epoxy thermoset	polymer	5.50	1.2	3.5	1.4	0.25	45	45	4.0	0.50	60
GFRP Laminate (glass)	composite	3.90	1.8	26	10	0.28	125	530	2.0	40	19
Glass (soda)	ceramic	1.35	2.5	65	26	0.23	3500	35	0.0	0.71	8.8
Granite	ceramic	3.15	2.6	66	26	0.25	2500	60	0.1	1.5	6.5
Ice (H_2O)	ceramic	0.23	0.92	9.1	3.6	0.28	85	6.5	0.0	0.11	55
Lead alloys	metal	1.20	11.1	16	5.5	0.45	33	42	60	40	29
Nickel alloys	metal	6.10	8.5	180	70	0.31	900	1200	30	93	13
Polyamide (nylon)	polymer	4.30	1.1	3.0	0.76	0.42	40	55	5.0	3.0	103
Polybutadiene elastomer	polymer	1.20	0.91	0.0016	0.0005	0.50	2.1	2.1	500	0.087	140
Polycarbonate	polymer	4.90	1.2	2.7	0.97	0.42	70	77	60	2.6	70
Polyester thermoset	polymer	3.00	1.3	3.5	1.4	0.25	50	0.7	2.0	0.70	150
Polyethylene (HDPE)	polymer	1.00	0.95	0.7	0.31	0.42	25	33	90	3.5	225
Polypropylene	polymer	1.10	0.89	0.9	0.42	0.42	35	45	90	3.0	85
Polyurethane elastomer	polymer	4.00	1.2	0.025	0.0086	0.50	30	30	500	0.30	125
Polyvinyl chloride (rigid PVC)	polymer	1.50	1.4	1.5	0.6	0.42	53	60	50	0.54	75
Silicon	ceramic	2.35	2.3	110	44	0.24	3200	35	0.0	1.5	6
Silicon Carbide (SiC)	ceramic	36.00	2.8	450	190	0.15	9800	35	0.0	4.2	4.2
Spruce (parallel to grain)	natural	1.00	0.60	9	0.8	0.30	48	50	10	2.5	4
Steel, high strength 4340	metal	0.25	7.8	210	76	0.29	1240	1550	2.5	100	14
Steel, mild 1020	metal	0.50	7.8	210	76	0.29	200	380	25	140	14
Steel, stainless austenitic 304	metal	2.70	7.8	210	76	0.28	240	590	60	50	17
Titanium alloy (6Al4V)	metal	16.25	4.5	100	39	0.36	910	950	15	85	9.4
Tungsten Carbide (WC)	ceramic	50.00	15.5	550	270	0.21	6800	35	0.0	3.7	5.8